ONDANSETRON

QT prolongation
Constipation

CHEMICAL NAME (RS)-9-methyl-3-[(2-methyl-1H-imidazol-1-yl)methyl]-2,3-dihydro-1H-carbazol-4(9H)-one (hydrochloride dihydrate)

CLASS 5-HT3 receptor antagonist antiemetic

TRADE NAMES Zofran · Emeset · Ondemet · Zofran Zydis (orodispersible)

THERAPEUTIC USES / INDICATIONS

- Chemotherapy- and radiotherapy-induced nausea and vomiting.
- Post-operative nausea and vomiting (PONV) — prevention and treatment.
- Gastroenteritis vomiting in children (single dose).
- Hyperemesis gravidarum (second-line).

ADMINISTRATION

- **IV** slowly over at least 30 s (single dose ≤8 mg) or infusion over 15 min (higher doses).
- **PO** tablets or orodispersible films/tablets (on tongue, no water).
- **IM**; **PR** suppository.
- Chemotherapy: 30 min before treatment.

DOSAGES

- **PONV:** 4 mg IV at end of surgery.
- **Chemotherapy:** 8 mg IV/PO before, then 8 mg every 12 h for up to 5 days; max single IV dose 16 mg (QT).
- **Children (gastroenteritis):** 0.1–0.15 mg/kg PO/IV once (max 4 mg).
- **Severe hepatic impairment:** max 8 mg/day.

PHARMACOKINETICS

- **Onset:** IV 30 min; PO 30–60 min. **Duration:** 8–12 h.
- Metabolised in liver (CYP3A4, 1A2, 2D6).
- Excreted in urine. **t½:** 3–6 h.

PHARMACODYNAMICS

- Blocks **5-HT3 receptors** on vagal afferents in gut and in chemoreceptor trigger zone (CTZ) → ↓ nausea and vomiting.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, constipation.
- Flushing, sensation of warmth.
- Dizziness.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, torsades de pointes.
- Serotonin syndrome (with serotonergic drugs).
- Hypersensitivity; transient visual disturbance (rapid IV).
- Possible small ↑ orofacial clefts if used in first trimester.

CONTRAINDICATIONS & CAUTIONS

- Congenital long QT syndrome.
- Use with apomorphine (profound hypotension).
- **Caution:** QT-prolonging drugs, electrolyte disturbance, heart failure, subacute intestinal obstruction.

KEY INTERACTIONS

- **QT drugs** (haloperidol, macrolides, citalopram, methadone): additive.
- **SSRIs, SNRIs, tramadol:** serotonin syndrome.
- **Apomorphine:** contraindicated.
- **Rifampicin, carbamazepine:** ↓ levels.

NURSING MANAGEMENT

- Check ECG/electrolytes in at-risk patients.
- Give before chemotherapy; schedule doses.
- Monitor vomiting frequency, fluid balance, bowel function.
- Correct K^+/Mg^{2+}.

PATIENT EDUCATION

- Place dissolving tablet on the tongue.
- Report palpitations or fainting.
- Increase fluids and fibre (constipation).
- Discuss use in early pregnancy with your doctor.

KEY POINTS

- 5-HT3 antagonist; chemo and PONV.
- Headache and constipation.
- Max 16 mg single IV dose (QT).
- Serotonin syndrome risk.

SOURCES CHECKED **[524]** Ondansetron 4 mg film-coated tablets (UK SmPC) **[525]** Ondansetron 2 mg/ml Solution for Injection (UK SmPC) **[526]** Ondansetron, IM, IV (US label) **[527]** Ondansetron, oral (US label)

Full references p. 432–457